Reyaa Divakar

608-810-0037 | Parker/Golden, CO

reyaadivakar_divakar@mines.edu | linkedin.com/in/reyaa-divakar | github.com/Cloudie-Rey

Education

Colorado School of Mines Aug 2026 – May 2029
Bachelor of Science in Mechanical Engineering, Concentration in Automation and Controls *Golden, CO*

Experience

Undergraduate Research Assistant Aug 2026 – Present
MIRRORLabs Colorado School of Mines *Golden, CO*

- Conduct physical robot body fabrication, structural assembly, and hardware testing for trauma-informed HRI research evaluating human psychological responses to robotic movement.
- Collaborate on physical fabrication and mechatronics assembly for a Fungi Learning Robot platform targeting submission to the HRI Student Design Competition as co-author.
- Execute hardware troubleshooting routines, log sensor calibration data, and assist in drafting technical research documentation for empirical lab trials.

Intern Jun 2025 – Jul 2025
Burns and McDonnell *Denver, CO*

- Participated in technical workshops, hardware fabrication sessions, and electrical soldering modules.
- Shadowed senior engineers across multidisciplinary plant design, controls infrastructure, and field engineering workflows.

Founder Aug 2023 – Aug 2026
Code Connect *Denver, CO*

- Founded and led a student-led nonprofit providing accessible computer science education through hands-on workshops, mentorship, curriculum development, and chapter growth.
- Taught 70+ hours of programming lessons to 80+ students covering Python, web development, and computer science fundamentals.

Projects

Mines Cornerstone: The Kraken Sound Sensor System | *Sensors, Microcontrollers, CAD* Aug 2026 – Present

- Co-engineering an acoustic search-and-rescue sensing platform to detect human voice frequencies under rubble for an upcoming Mines design competition.
- Integrating Arduino acoustic sensors and signal-filtering pipelines to isolate vocal frequencies from environmental noise during emergency operations.

LuckyCharm AI Vision Assistant | *Python, OpenCV, Raspberry Pi, Computer Vision* Jun 2026 – Present

- Developing an AI-powered vision assistant that analyzes real-time camera input, highlights relevant information, and provides solutions based on user commands.

Technical Skills

Programming: MATLAB, Python, Java, HTML/CSS, JavaScript, C#
Hardware & Tools: Arduino, Raspberry Pi, Sensors, Multimeters, Soldering, Circuit Assembly
CAD & Software: SolidWorks, OnShape, OpenCV, MATLAB, Unity, Vercel, Git, Figma
Languages: English (Bilingual), Tamil (Bilingual), Chinese (Limited Working), Spanish (Elementary)

Awards

Outstanding Technology and Engineering Student of the Year May 2026
Rock Canyon High School *Highlands Ranch, CO*

- Departmental award recognizing one outstanding student per grade.

3rd Place in State for Engineering CAD March 2026
Science Olympiad Colorado *Colorado Springs, CO*

- Placed 4th at the district competition and 3rd at the State Science Olympiad Conference, where entries were evaluated based on model quality and completion speed.